SCRIPT YA VIDEO — SIMPLE CALCULATOR (JAVA)

🎬 SEHEMU YA 1: MWANZO (INTRO)

Karibu tena kwenye channel yetu!

Leo tutajifunza jinsi ya kutengeneza **Simple Calculator** kwa lugha ya **Java**. Hii ni project nzuri sana kwa wale ambao ni **beginners** kwenye programming, kwa sababu itakusaidia kuelewa mambo ya msingi kama:

- Variables
- Loops (while)
- Conditionals (if na switch)
- Kupokea input kutoka kwa mtumiaji kwa kutumia Scanner

Kabla hatujaanza, hakikisha una **JDK** (Java Development Kit) imewekwa kwenye computer yako, na una editor kama VS Code au IntelliJ.

Sasa twende kazini!

🎬 SEHEMU YA 2: KUANGALIA CODE YOTE

Kwanza, hebu tuangalie code nzima ili tupate picha kamili, kisha tutaivunja vunja part by part.

```java
import java.util.Scanner;

public class Calculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        boolean running = true;

        System.out.println("===== SIMPLE CALCULATOR =====");

        while (running) {
            System.out.println("\nChoose operation:");
            System.out.println("1. Add");
            System.out.println("2. Subtract");
            System.out.println("3. Multiply");
            System.out.println("4. Divide");
            System.out.println("5. Exit");
            System.out.print("Enter choice (1-5): ");
            int choice = sc.nextInt();

            if (choice == 5) {
                running = false;
                System.out.println("Goodbye!");
                continue;
            }

            System.out.print("Enter first number: ");
            double num1 = sc.nextDouble();
            System.out.print("Enter second number: ");
            double num2 = sc.nextDouble();
            double result;

            switch (choice) {
                case 1:
                    result = num1 + num2;
                    System.out.println("Result: " + result);
                    break;
                case 2:
                    result = num1 - num2;
                    System.out.println("Result: " + result);
                    break;
                case 3:
                    result = num1 * num2;
                    System.out.println("Result: " + result);
                    break;
                case 4:
                    if (num2 == 0) {
                        System.out.println("Error: Cannot divide by zero.");
                    } else {
                        result = num1 / num2;
                        System.out.println("Result: " + result);
                    }
                    break;
```

```
                default:
                    System.out.println("Invalid choice. Try again.");
            }
        }
        sc.close();
    }
}
```

📻 SEHEMU YA 3: KUELEZA CODE LINE BY LINE

Sasa hebu tuivunje vunje, sehemu kwa sehemu.

3.1 — Import na Class

```
import java.util.Scanner;

public class Calculator {
```

Hapa tunaita (import) class ya `Scanner`, ambayo inatusaidia **kupokea input** kutoka kwa mtumiaji — kwa mfano namba, au maneno anayoandika kwenye terminal.

Kisha tunatengeneza class yetu kuu inayoitwa `Calculator`. Kila Java program lazima iwe na class, na jina la file lazima liwe sawa na jina la class — kwa hiyo file hii inaitwa `Calculator.java`.

3.2 — Method Main

```
public static void main(String[] args) {
```

Hii ndiyo **method kuu** — sehemu ambayo program inaanzia kukimbia (execute). Kila program ya Java inatakiwa iwe na `main` method.

3.3 — Kutengeneza Scanner na Variable ya Kudhibiti Loop

```
Scanner sc = new Scanner(System.in);
boolean running = true;
```

Hapa: - Tunatengeneza object ya `Scanner` inayoitwa `sc`, ambayo tutaitumia kusoma kile mtumiaji anachoandika. - Tunatengeneza variable ya `boolean` inayoitwa `running`, na tunaipa thamani `true`. Variable hii ndiyo itakayodhibiti mzunguko (loop) wa calculator — maadamu ni `true`, calculator itaendelea kufanya kazi.

3.4 — Loop Kuu (While Loop)

```
while (running) {
```

Hii ni **while loop**. Loop hii itaendelea kukimbia mradi tu `running` ni `true`. Ndani ya loop hii ndipo calculator yetu inafanya kazi zake zote — kuonyesha menu, kupokea choice, kufanya hesabu, na kurudia tena.

3.5 — Kuonyesha Menu

```
System.out.println("\nChoose operation:");
System.out.println("1. Add");
System.out.println("2. Subtract");
System.out.println("3. Multiply");
System.out.println("4. Divide");
System.out.println("5. Exit");
System.out.print("Enter choice (1-5): ");
int choice = sc.nextInt();
```

Hapa tunamwonyesha mtumiaji chaguo tano: Kujumlisha, Kutoa, Kuzidisha, Kugawanya, au Kutoka (Exit). Kisha tunatumia `sc.nextInt()` kusoma namba aliyoandika, na kuihifadhi kwenye variable `choice`.

3.6 — Kutoka Kwenye Program (Exit)

```
if (choice == 5) {
    running = false;
    System.out.println("Goodbye!");
    continue;
}
```

Kama mtumiaji ameandika `5`, tunabadilisha `running` kuwa `false` — hii itasababisha loop kusimama kwenye mzunguko unaofuata. Neno `continue` linaruka moja kwa moja mpaka mwanzo wa loop, ili isiendelee kuuliza namba za hesabu.

3.7 — Kuchukua Namba Mbili

```
System.out.print("Enter first number: ");
double num1 = sc.nextDouble();
```

```
        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();
        double result;
```

Hapa tunamuomba mtumiaji aandike namba mbili — `num1` na `num2` — kama `double`, ili ziweze kuwa na desimali pia (mfano 4.5). Pia tunatengeneza variable `result` itakayohifadhi jibu la hesabu.

3.8 — Switch Statement (Kufanya Hesabu)

```
        switch (choice) {
            case 1:
                result = num1 + num2;
                System.out.println("Result: " + result);
                break;
```

`switch` inaangalia thamani ya `choice`, na kutegemea na namba aliyochagua, inafanya operesheni husika:

- **case 1** → Kujumlisha (`+`)
- **case 2** → Kutoa (`-`)
- **case 3** → Kuzidisha (`*`)
- **case 4** → Kugawanya (`/`) — hapa tunaongeza ulinzi maalum:

```
            case 4:
                if (num2 == 0) {
                    System.out.println("Error: Cannot divide by zero.");
                } else {
                    result = num1 / num2;
                    System.out.println("Result: " + result);
                }
                break;
```

Hii ni muhimu sana — kwenye hesabu, huwezi kugawanya namba na sifuri (zero). Kwa hiyo tunaangalia kwanza kama `num2` ni sifuri; kama ndiyo, tunaonyesha ujumbe wa error badala ya kusababisha program iharibike (crash).

Mwishoni tunaweka:

```
            default:
                System.out.println("Invalid choice. Try again.");
```

Hii inahakikisha kama mtumiaji ameandika namba isiyokuwepo kwenye menu (mfano 9), program haitaharibika — itamwambia tu chaguo si sahihi.

3.9 — Kufunga Scanner

```
        sc.close();
```

Baada ya loop kuisha (yaani `running` imekuwa `false`), tunafunga `Scanner` object. Hii ni tabia nzuri ya programming — kufunga resources ambazo hazihitajiki tena.

📻 SEHEMU YA 4: DEMO — KUENDESHA PROGRAM

Sasa hebu tuone jinsi program hii inavyofanya kazi tukiiendesha (run).

Hatua za kuendesha:

1. Fungua Terminal au Command Prompt.
2. Nenda kwenye folder lililo na file `Calculator.java`.
3. Andika: `javac Calculator.java` — hii ita-compile code yetu.
4. Andika: `java Calculator` — hii itaendesha program.

(Hapa unaonyesha screen recording ukijaribu chaguo tofauti — kujumlisha, kutoa, na hatimaye kujaribu kugawanya na sifuri ili kuonyesha error message inavyofanya kazi.)

📻 SEHEMU YA 5: HITIMISHO (OUTRO)

Na hapo ndipo tunamalizia Simple Calculator yetu kwa Java!

Kwa muhtasari, umejifunza:

✅ Jinsi ya kutumia `Scanner` kupokea input ✅ Jinsi `while loop` inavyoweza kuendesha program mpaka mtumiaji atoke ✅ Jinsi `switch statement` inavyosaidia kuchagua kati ya chaguo tofauti ✅ Jinsi ya kuzuia error ya kugawanya na sifuri

Kama video hii imekusaidia, usisahau **kulike**, **kucomment** kama una swali, na **kusubscribe** channel yetu ili usikose video nyingine za programming.

Kwenye video ijayo, tutatengeneza project nyingine — labda **Number Guessing Game** au **To-Do List App**. Niambie kwenye comments unataka nifanyie project ipi ifuatayo!

Asanteni sana kwa kunifuatilia, tuonane video ijayo. Kwaherini!

(MWISHO WA SCRIPT)